

Database Systems and Application Development

Lab # 4

Fall 2026

**Constraints, Primary Keys, Foreign Keys,
Data Types.**

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- Exercise 1: Creating a Table

Create a table named students with the following columns: ...g

```
[exercises4=# create table students(  
  student_id int primary key,  
  first_name varchar(50),  
  last_name varchar(50),  
  age int,  
  grade varchar(2));  
CREATE TABLE
```

```
[exercises4=# \dt
```

List of tables

Schema	Name	Type	Owner
public	students	table	250123

(1 row)

- Exercise 2: Inserting Data

Insert 10 records into the students table you created in Exercise 1.

```
exercises4=# insert into students (student_id, first_name, last_name, age, grade)
exercises4=# VALUES
exercises4=# (1, 'stud1', 'lstud1', 18, '12'),
exercises4=# (2, 'stud2', 'lstud2', 18, '12'),
exercises4=# (3, 'stud3', 'lstud3', 18, '12'),
exercises4=# (4, 'stud4', 'lstud4', 18, '12'),
exercises4=# (5, 'stud5', 'lstud5', 18, '12'),
exercises4=# (6, 'stud6', 'lstud6', 18, '12'),
exercises4=# (7, 'stud7', 'lstud7', 18, '12'),
exercises4=# (8, 'stud8', 'lstud8', 18, '12'),
exercises4=# (9, 'stud9', 'lstud9', 18, '12'),
exercises4=# (10, 'stud10', 'lstud10', 18, '12');
INSERT 0 10
```

- Exercise 3: Creating another Table
Create a table named courses with the following columns: ...

```
exercises4=# create table courses(  
course_id int primary key,  
course_name varchar(100),  
credits int);  
CREATE TABLE
```

- Exercise 4: Inserting Data into the Courses Table
Insert four records into the courses table you created in Exercise 3.

```
exercises4=# insert into courses(course_id, course_name, credits)
values
(1, 'course1', 5),
(2, 'course2', 6),
(3, 'course3', 5),
(4, 'course4', 4);
INSERT 0 4
```

- Exercise 5: Creating a Table with Foreign Key Constraint
Create a table named enrollments with the following columns:

```
exercises4=# create table enrollments(  
  enrollment_id int primary key,  
  student_id int,  
  course_id int,  
  enrollment_date date,  
  constraint stid foreign key (student_id) references students(student_id),  
exercises4=# constraint coid foreign key (course_id) references courses(course_id));  
CREATE TABLE
```

Exercise 6: Inserting Data into the Enrollments Table

Insert five records into the enrollments table you created in Exercise 5.

```
exercises4=# insert into enrollments (enrollment_id, student_id, course_id, enrollment_date)
exercises4=# values
exercises4-# (1, 1, 1, '2026-09-01'),
exercises4-# (2, 2, 1, '2026-09-01'),
exercises4-# (3, 3, 2, '2026-09-02'),
exercises4-# (4, 4, 3, '2026-09-03'),
exercises4-# (5, 5, 4, '2026-09-04');
INSERT 0 5
```

- Exercise 7: Creating a Table with Default Values
Create a table named employees with the following columns:

```
[exercises4=# create table employees(  
[exercises4(# employee_id int primary key,  
[exercises4(# first_name varchar(50),  
[exercises4(# last_name varchar(50),  
[exercises4(# hire_date date default current_date,  
[exercises4(# department varchar(50) default 'Unknown');  
CREATE TABLE
```


- Exercise 8: Inserting Data into the Employees Table
Insert six records into the employees table you created in Exercise 7.

```
exercises4=# insert into employees (employee_id, first_name, last_name)
exercises4=# values
exercises4=# (1, 'John', 'Doe'),
exercises4=# (2, 'Jane', 'Smith'),
exercises4=# (3, 'Alex', 'Jones'),
exercises4=# (4, 'Emily', 'Brown'),
exercises4=# (5, 'Michael', 'Green'),
exercises4=# (6, 'Sarah', 'White');
INSERT 0 6
```

```
exercises4=# \dt
```

List of tables

Schema	Name	Type	Owner
public	courses	table	250123
public	employees	table	250123
public	enrollments	table	250123
public	students	table	250123

(4 rows)

```
exercises4=# select * from employees
```

```
exercises4=# ;
```

employee_id	first_name	last_name	hire_date	department
1	John	Doe	2026-09-23	Unknown
2	Jane	Smith	2026-09-23	Unknown
3	Alex	Jones	2026-09-23	Unknown
4	Emily	Brown	2026-09-23	Unknown
5	Michael	Green	2026-09-23	Unknown
6	Sarah	White	2026-09-23	Unknown

(6 rows)

- Exercise 9: Updating Records

Assuming you have the students table from the previous exercises:

```
[exercises4=# update students set age = 21 where student_id = 1;
UPDATE 1
[exercises4=# update students set grade = 'B' where student_id = 2;
UPDATE 1
[exercises4=# update students set last_name = 'Smith' where student_id = 3;
UPDATE 1
[exercises4=# select * from students
[exercises4=# ;
```

student_id	first_name	last_name	age	grade
4	stud4	lstud4	18	12
5	stud5	lstud5	18	12
6	stud6	lstud6	18	12
7	stud7	lstud7	18	12
8	stud8	lstud8	18	12
9	stud9	lstud9	18	12
10	stud10	lstud10	18	12
1	stud1	lstud1	21	12
2	stud2	lstud2	18	B
3	stud3	Smith	18	12

(10 rows)

- Exercise 10: Updating Records Based on a Condition
Assuming you have the courses table from the previous exercises:

```
[exercises4=# update courses set credits = credits+1;  
UPDATE 4  
[exercises4=# update courses set course_name = 'Advanced Mathematics' where course_id = 2;  
UPDATE 1
```

- Exercise 11: Deleting Records
Assuming you have the employees table from the previous exercises

```
[exercises4=# update courses set credits = credits+1;
UPDATE 4
[exercises4=# update courses set course_name = 'Advanced Mathematics' where course_id = 2;
UPDATE 1
[exercises4=# delete from employees where employee_id = 4;
DELETE 1
```

```
[exercises4=# delete from employees where hire_date < '2023-01-01';
DELETE 0
```

```
[exercises4=# select * from employees
[exercises4=# ;
 employee_id | first_name | last_name | hire_date | department
-----+-----+-----+-----+-----
          1 | John      | Doe       | 2026-09-23 | Unknown
          2 | Jane      | Smith     | 2026-09-23 | Unknown
          3 | Alex      | Jones     | 2026-09-23 | Unknown
          5 | Michael   | Green     | 2026-09-23 | Unknown
          6 | Sarah     | White     | 2026-09-23 | Unknown
(5 rows)
```